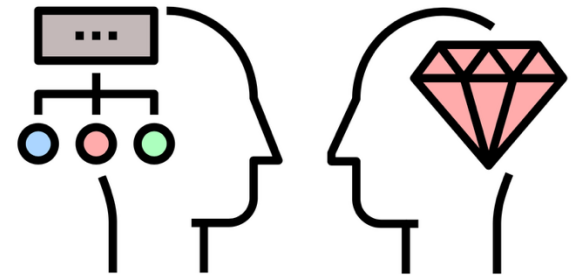


Machine Learning for Materials

4. Crystal Representations

Aron Walsh

Department of Materials
Centre for Processable Electronics

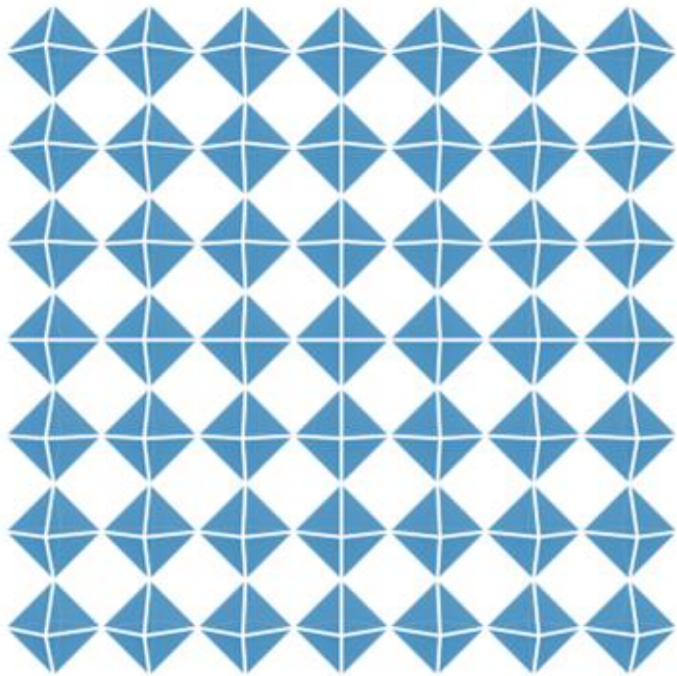


Module Contents

1. Introduction
 2. Machine Learning Basics
 3. Materials Data
 - 4. Crystal Representations**
 5. Classical Learning
 6. Deep Learning
 7. Building a Model from Scratch
 8. Accelerated Discovery
 9. Generative Artificial Intelligence
 10. Future Directions
-

Key Concept #4

Material (M)



**Numerical
representation (X)**

$$X = \phi(M) \rightarrow \begin{bmatrix} x_{ii} & \cdots & x_{ij} \\ \vdots & \ddots & \vdots \\ x_{ji} & \cdots & x_{jj} \end{bmatrix}$$

Class Outline

Crystal Representations

A. Compositional

B. Structural

C. Graphs

Representation of Materials

Model performance depends on the choice of compositional and structural features

Minimal representation

Ab initio quantum mechanics (QM)

Input:

Atomic number, Z

Coordinates, R

$$\hat{H}|\Psi\rangle = E|\Psi\rangle$$

electronic
wavefunction

Output:

Properties

Effective representation

Supervised machine learning (ML)

Input:

Feature vector, X

$$y = f(X, \Theta)$$

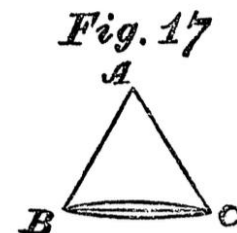
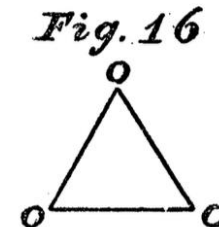
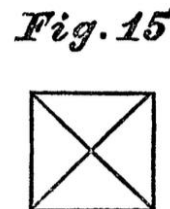
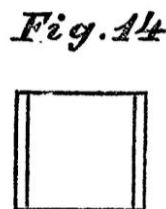
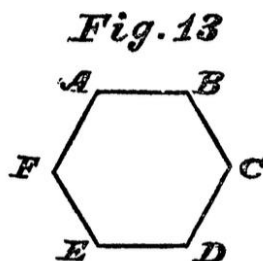
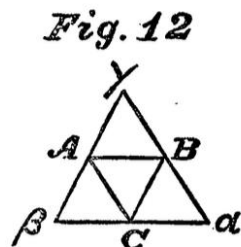
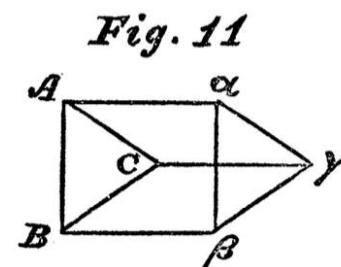
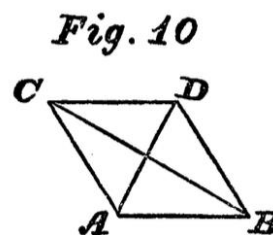
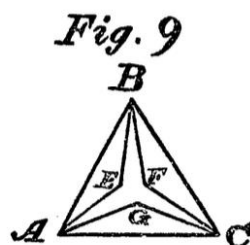
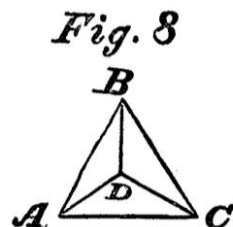
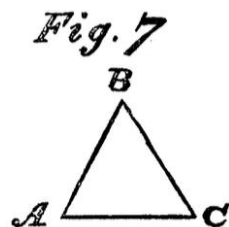
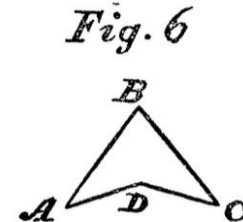
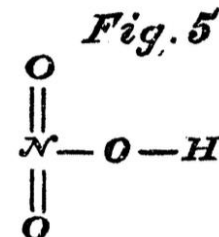
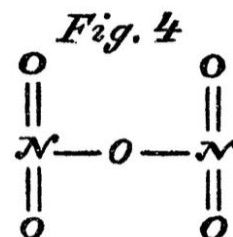
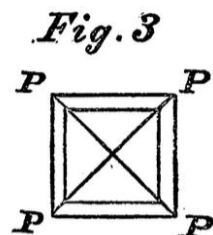
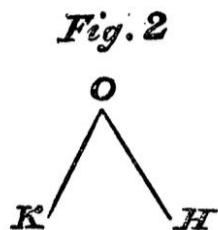
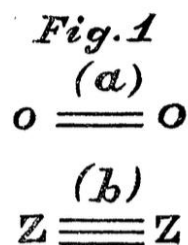
learned
weights

Output:

Properties

How to Best Represent a Molecule?

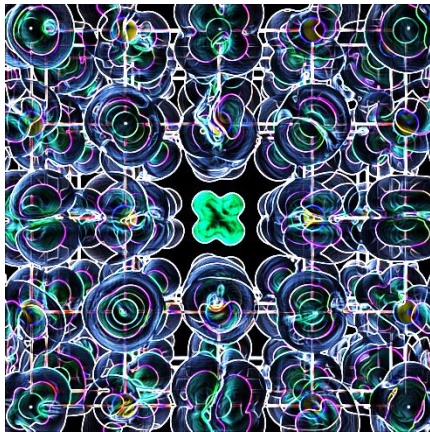
Networks of atoms (nodes) connected by bonds (edges)



How to Best Represent a Material?

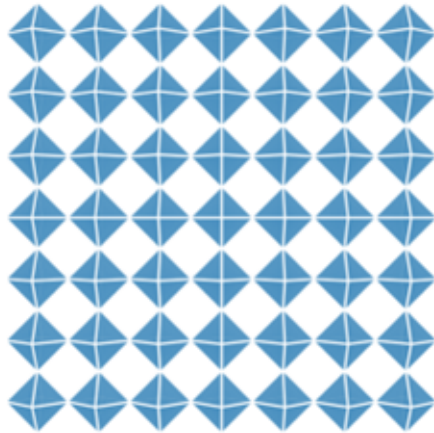
Wide range of possible materials features
from atomistic to macroscopic length scales

Electronic



Wavefunctions
or electron density
(Å)

Nanoscale



Short-range
connectivity
(nm)

Microstructure



Grain size, shape,
orientation
(μm)

Macroscale



Continuum
properties
(cm)

Hot Encoding

We can use an n -dimensional vector to categorise the atomic number of the elements in a compound

Element (One-hot)

[1 0 0 0 0 0 0 0 0 ...]

H He Li Be B C N O F....

Compound (Multi-hot)

[0 0 0 0 0 1 0 1 0 ...]

H He Li Be B C N O F....

'1' indicates the presence of that specific element and '0' for others

Hand-Built (Local) Representations

We can *define* elemental feature vectors based on standard properties of the elements

```
import elementembeddings

print(AtomEmbeds["magpie"].dim)

22

print(AtomEmbeds["magpie"].feature_labels)

['Number', 'MendeleevNumber', 'AtomicWeight', 'MeltingT', 'Column', 'Row', 'CovalentRadius', 'Electronegativity', 'NsValence', 'NpValence', 'NdValence', 'NfValence', 'NValence', 'NsUnfilled', 'NpUnfilled', 'NdUnfilled', 'NfUnfilled', 'NUnfilled', 'GSvolume_pa', 'GSbandgap', 'GSmagmom', 'SpaceGroupNumber']
```

22 dimensional Magpie representation from
L. Ward et al, npj Comp. Mater. 2, 16028 (2016)

Hand-Built (Local) Representations

We can also *define* compound feature vectors based on standard properties of the elements

```
Fe2O3_magpie = CompositionalEmbedding("Fe2O3", "magpie")
```

$$\mathbf{X}^{(\text{Fe}_2\text{O}_3)} = [2\mathbf{X}^{(\text{Fe})} + 3\mathbf{X}^{(\text{O})}]/5$$

	X_1	X_2	X_3	... X_n
Fe	0.52	0.11	0.01	0.80
O	0.32	0.23	0.14	0.64
Fe ₂ O ₃	0.40	0.18	0.09	0.70

Different types of pooling is possible (e.g. max, min, mean)

Learned (Distributed) Representations

We can *learn* continuous feature vectors with elemental information as part of model training

SkipSpecies

200 D

Structure
graph pooling

APL Machine Learning

Ionic species representations for materials informatics

Cite as: APL Mach. Learn. 2, 036112 (2024); doi: [10.1063/5.0227009](https://doi.org/10.1063/5.0227009)

Submitted: 5 July 2024 • Accepted: 28 August 2024 •

Published Online: 19 September 2024



View Online



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CrossMark

Anthony Onwuli,¹  Keith T. Butler,^{2,a)}  and Aron Walsh^{1,b)} 

LETTER

<https://doi.org/10.1038/s41586-019-1335-8>

Unsupervised word embeddings capture latent knowledge from materials science literature

Vahe Tshitoyan^{1,3*}, John Dagdelen^{1,2}, Leigh Weston¹, Alexander Dunn^{1,2}, Ziqin Rong¹, Olga Kononova², Kristin A. Persson^{1,2}, Gerbrand Ceder^{1,2*} & Anubhav Jain^{1*}

Element Embeddings

Toolkit to access and modify elemental and compositional representations for machine learning

ElementEmbeddings

Made with Python License MIT code style black open issues 6 ElementEmbeddings CI passing
codecov 71% DOI 10.5281/zenodo.8117601 pypi v0.1.1 docs mkdocs material python 3.8 | 3.9 | 3.10

The **Element Embeddings** package provides high-level tools for analysing elemental embeddings data. This primarily involves visualising the correlation between embedding schemes using different statistical measures.

Motivation

Machine learning approaches for materials informatics have become increasingly widespread. Some of these involve the use of deep learning techniques where the representation of the elements is learned rather than specified by the user of the model. While an important goal of machine learning training is to minimise the chosen error function to make more accurate predictions, it is also important for us material scientists to be able to interpret these models. As such, we aim to evaluate and compare different atomic embedding schemes in a consistent framework.

Getting started

ElementEmbeddings's main feature, the Embedding class is accessible by importing the class.



Dr Anthony Onwuli
(Matnex)

Latest embeddings

CrystaLLM

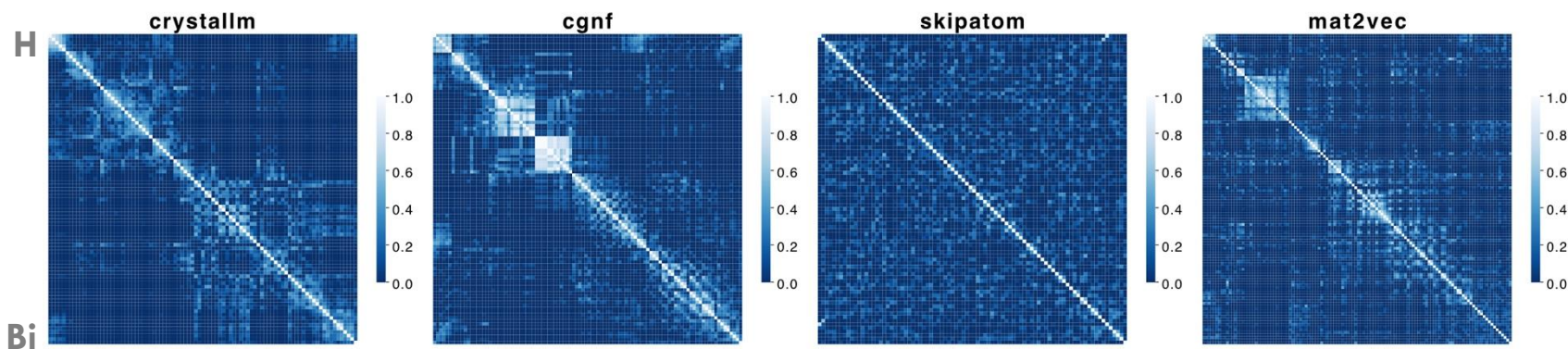
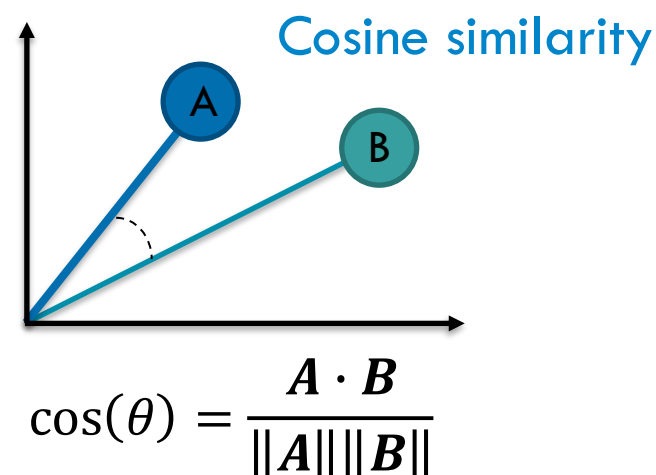
SkipSpecies

CGNF

Learned Chemical Similarity

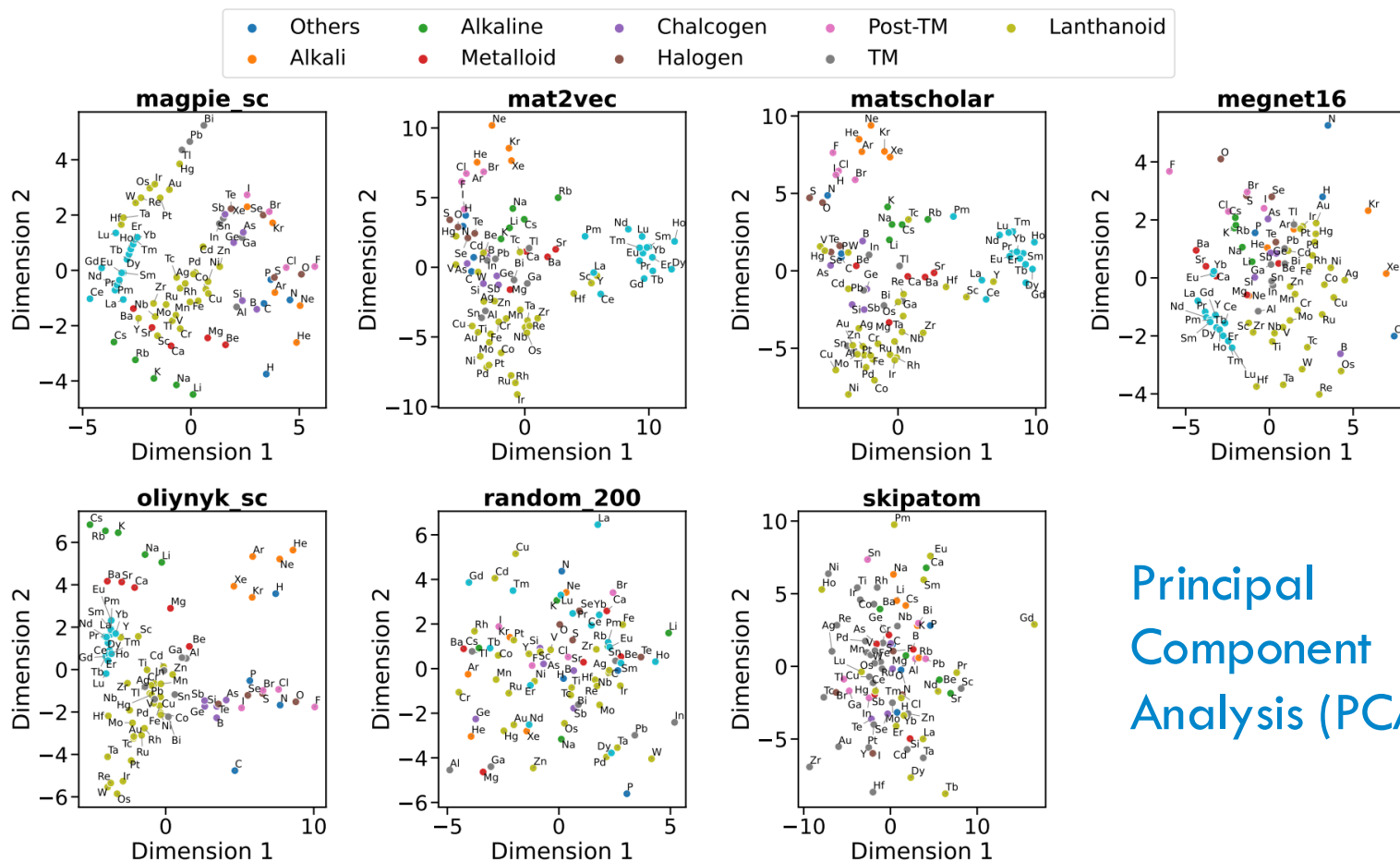
Quantify with distance (e.g. Chebyshev), similarity (e.g. Cosine), or correlation (e.g. Pearson) metrics

Name	Dimension	Type
Magpie	22	Element properties
Mat2Vec	200	Chemical abstracts
Skipatom	200	Crystal structure graphs
MegNet	16	Graph neural network
CrystaLLM	512	Crystal structure text



Learned Chemical Similarity

Dimensionality reduction confirms a natural clustering of elements into “groups”



Principal
Component
Analysis (PCA)

Class Outline

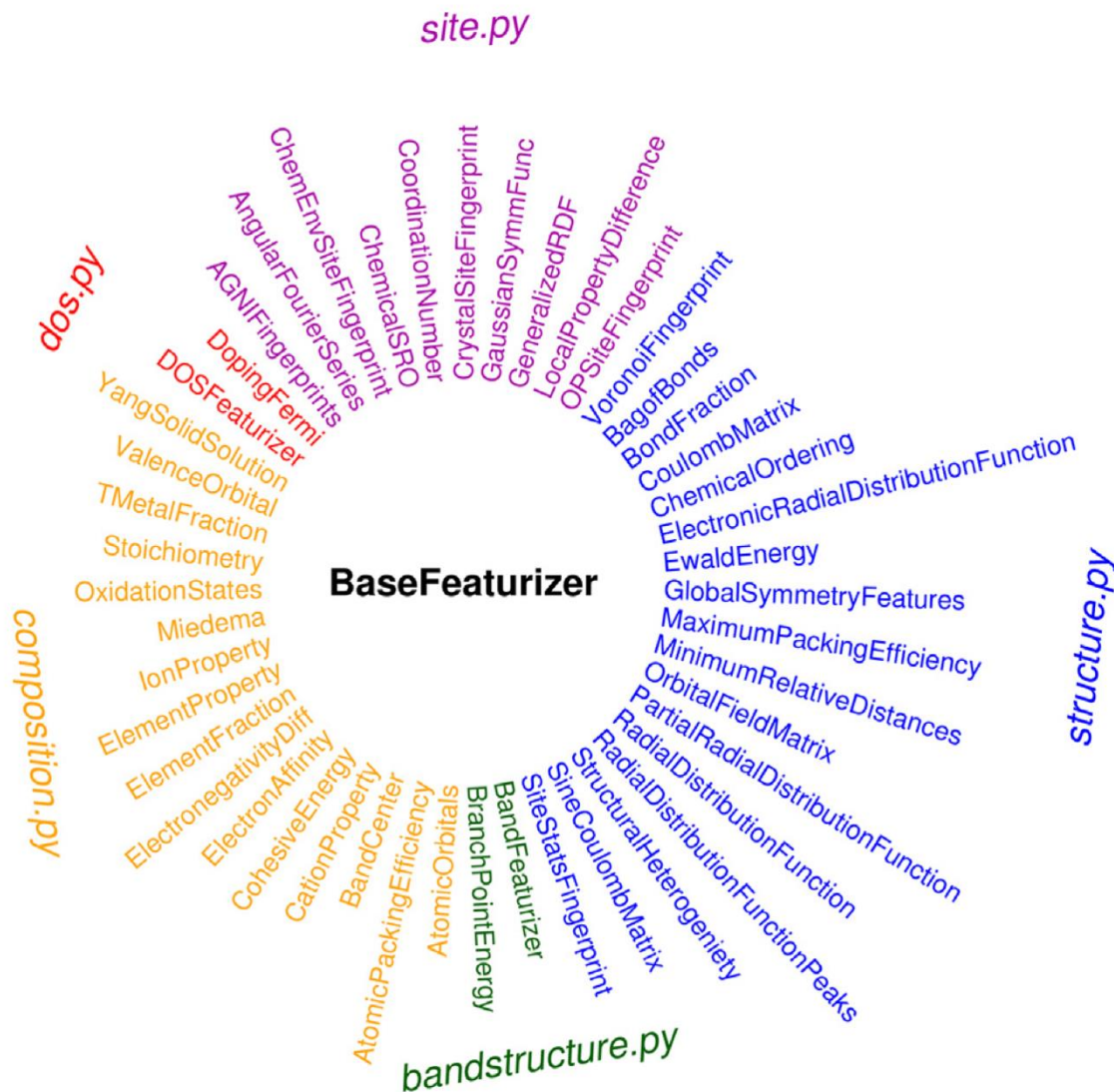
Crystal Representations

A. *Compositional*

B. *Structural*

C. *Graphs*

Many Possible Materials Features



Learn from Crystallography

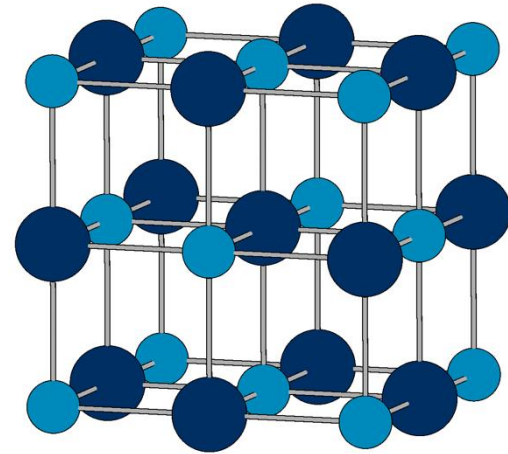
High symmetry crystal:



Cubic

8 atom unit cell

$$a = b = c$$



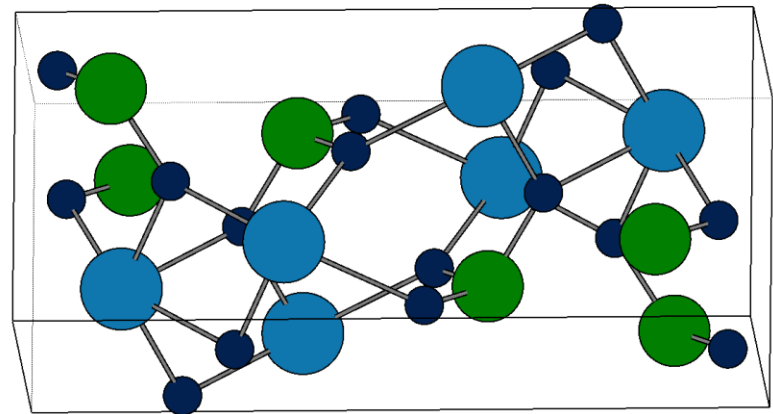
Low symmetry crystal:



Monoclinic

24 atom unit cell

$$a \neq b \neq c$$



Learn from Crystallography

7 crystal systems, 14 Bravais lattices,
230 space groups, 10^3 prototype structures

Conventional description

Unit cell ($\mathcal{L}$)

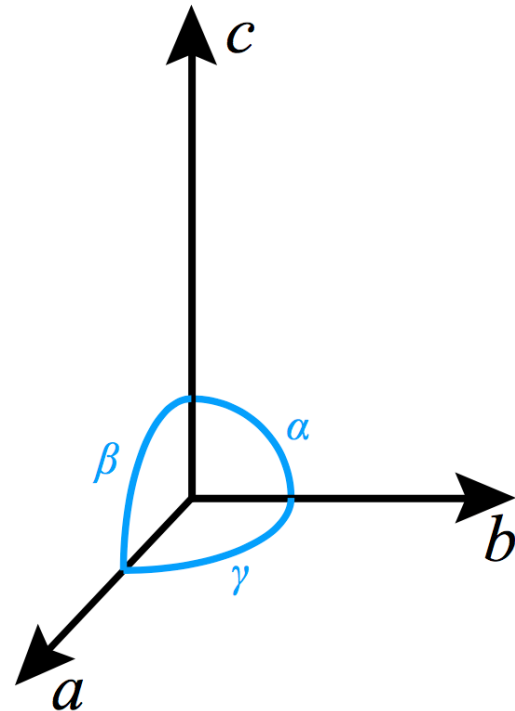
$a, b, c, \alpha, \beta, \gamma$

Fractional coordinates ($\mathcal{X}$)

$(x_1, y_1, z_1) \dots$

Atom types ($\mathcal{A}$)

Sn, Ti, O...



*Problem for ML: conventional description lacks invariance**

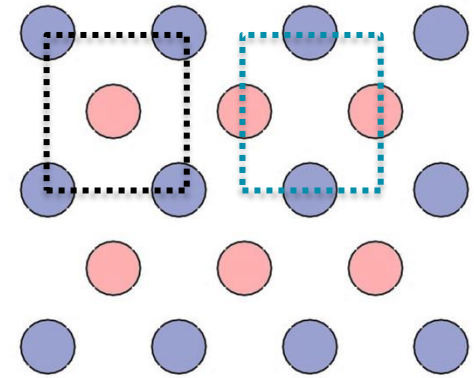
*with respect to atomic permutation, unit cell rotations, and translations

Unit Cell Transformations

The same structure is described in each case

Two-atom orthorhombic unit cell

$$\begin{bmatrix} a & b & c \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{bmatrix} \begin{bmatrix} 4 & 5 & 6 \\ 0 & 0 & 0 \\ 0.5 & 0.5 & 0.5 \end{bmatrix}$$



Atomic permutation

$$\begin{bmatrix} 4 & 5 & 6 \\ 0.5 & 0.5 & 0.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Crystal rotation

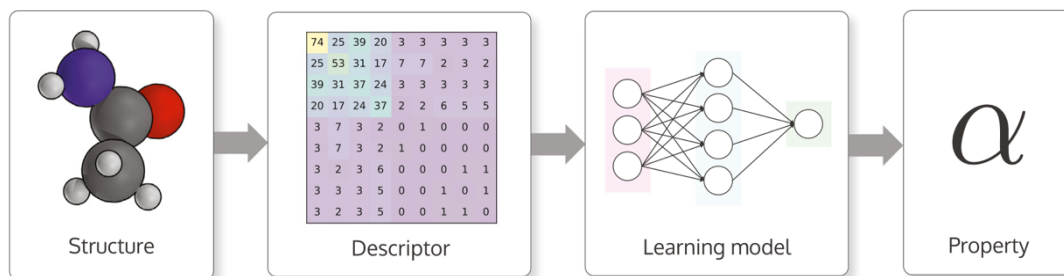
$$\begin{bmatrix} 5 & 4 & 6 \\ 0.5 & 0.5 & 0.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Unit cell translation

$$\begin{bmatrix} 4 & 5 & 6 \\ 0.0 & 0.5 & 0.5 \\ 0.5 & 0 & 0 \end{bmatrix}$$

Structural Representations

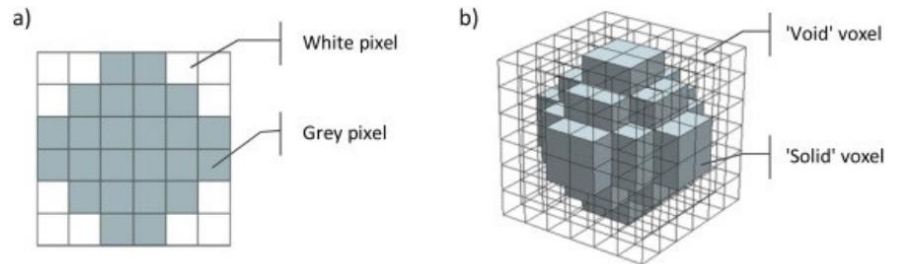
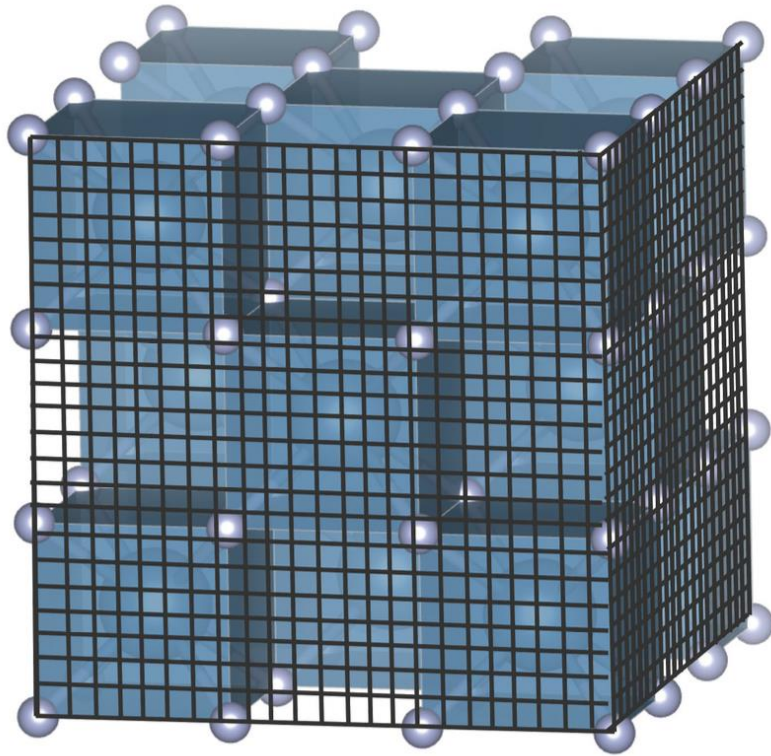
Many structural descriptors have been developed



- **Coulomb Matrix** (Rupp et al, 2012)
 - mimics electrostatic interactions ($q_i q_j / r_{ij}$)
- **Atom-Centered Symmetry Functions** (Behler, 2011)
 - site expansion of radial and angular terms
- **Many Body Tensor Representation** (Huo et al, 2017)
 - distribution of local structural motifs
- **Atomic Cluster Expansion** (Drautz, 2019)
 - high body-order expansion of atomic environments

Real Space Grid

Voxels (three-dimensional pixels) used in computer graphics can describe a unit cell

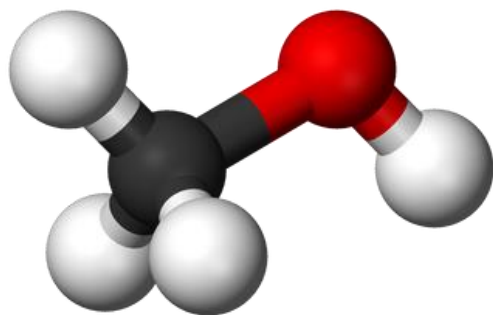


Used in early materials ML, but not recommended for structure

Pairwise Interatomic Distances

Coulomb matrix is a global descriptor that mimics the electrostatic interaction between nuclei

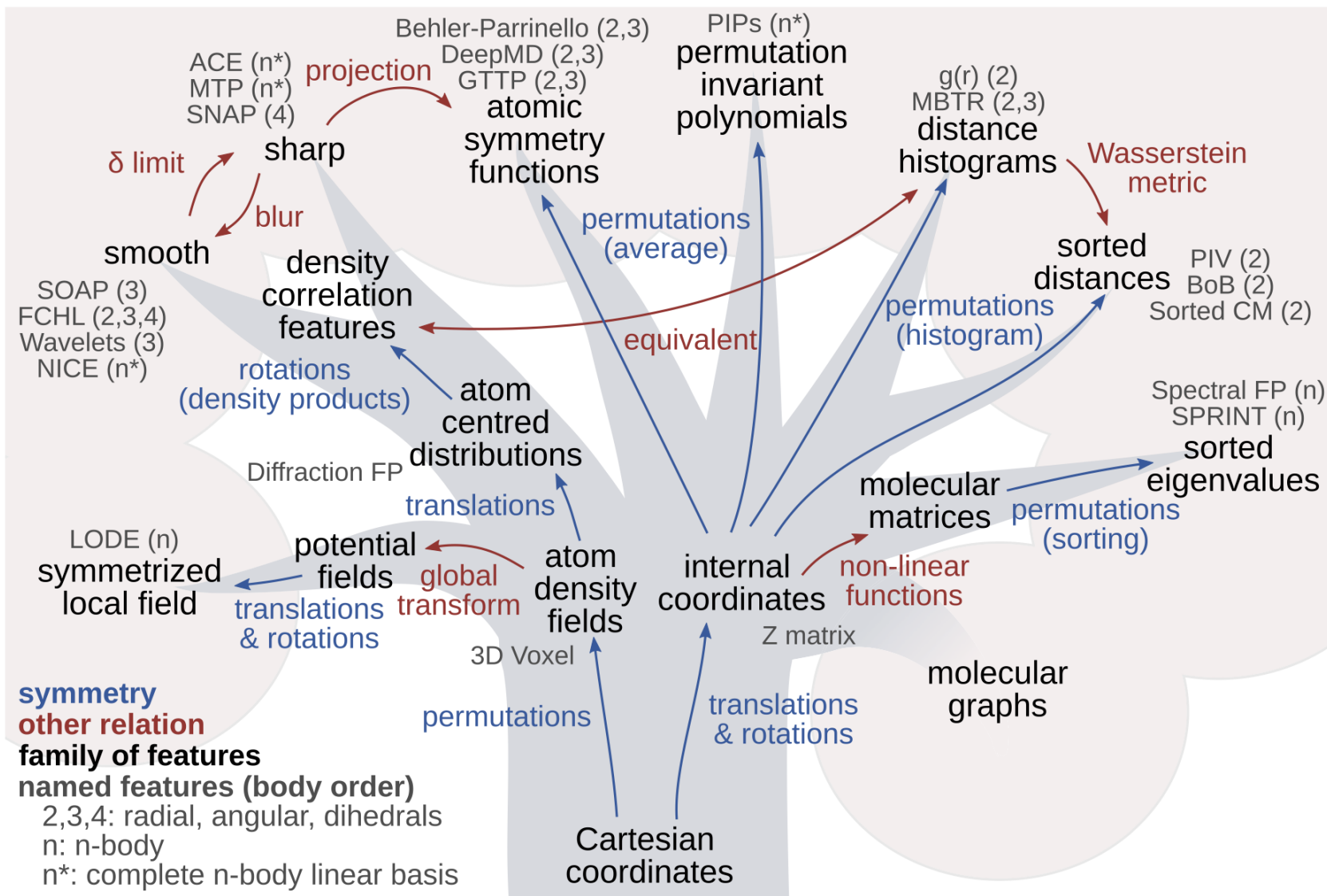
$$M_{ij}^{\text{Coulomb}} = \begin{cases} 0.5Z_i^{2.4} & \text{for } i = j \\ \frac{Z_i Z_j}{R_{ij}} & \text{for } i \neq j \end{cases}$$



36.9	33.7	5.5	3.1	5.5	5.5
33.7	73.5	4.0	8.2	3.8	3.8
5.5	4.0	0.5	0.35	0.56	0.56
3.1	8.2	0.35	0.5	0.43	0.43
5.5	3.8	0.56	0.43	0.5	0.56
5.5	3.8	0.56	0.43	0.56	0.5

Sine matrix is a modification that accounts for periodicity

Invariant Structural Representations



Invariant Structural Representations

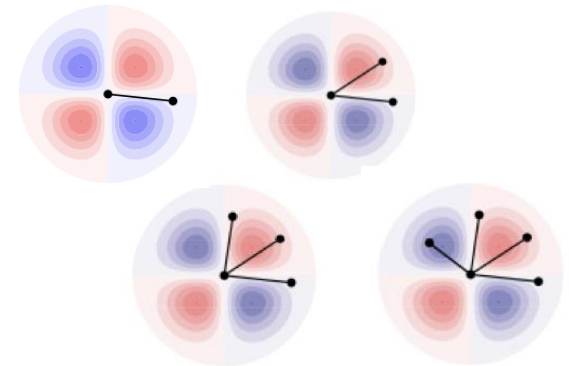
Atomic Cluster Expansion (ACE) provides a systematic representation of atomic environments through radial (R) and angular (Y) terms

Site basis function $\phi(r) = R_l Y_l^m$

Permutation invariance $A_i = \sum_{\text{neighbours}} \phi(r)$

Rotation (Q) invariance $B_i = \int A_i dQ$

Product basis B
forms a body-order
expansion

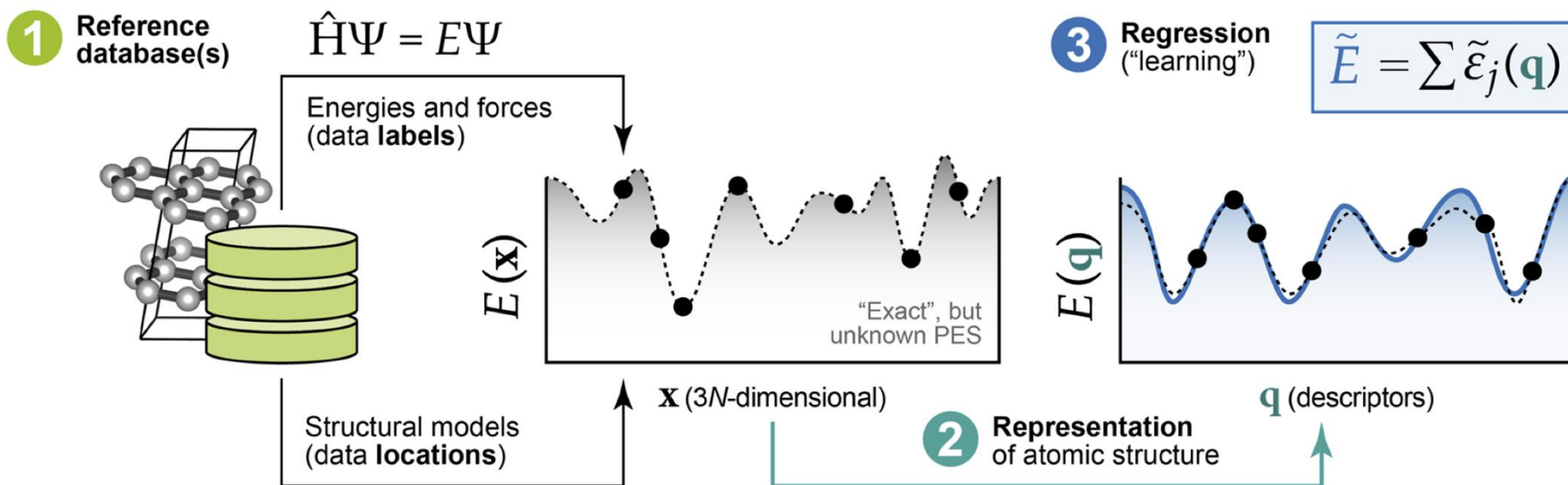


Property = $f(\underset{\text{weights}}{B}, \Theta)$

ACE is used in linear and
deep learning models for materials

ML Powered Molecular Dynamics

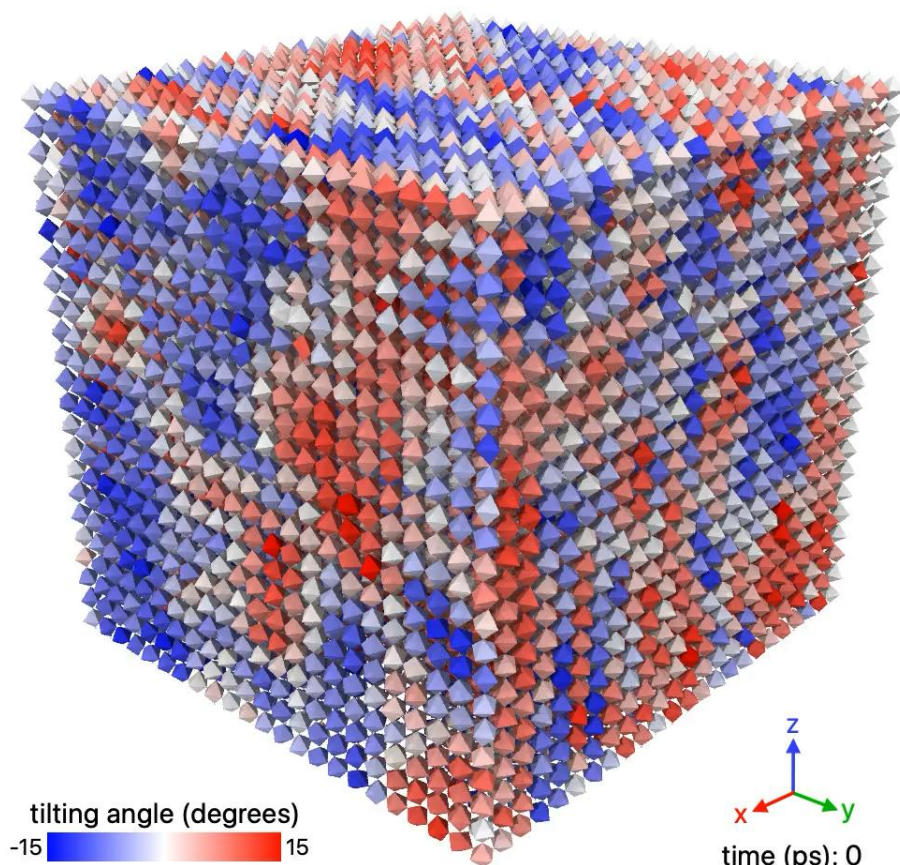
Classical models are being complemented by machine learning force fields (MLFF)



Three start-of-the-art implementations based on equivariant neural network regression are MACE, Allegro, and SevenNet

ML Powered Molecular Dynamics

Enable large-scale simulations of complex materials such as organic-inorganic solids



69,120 atom
simulation of CsPbI_3
perovskite based on
the atomic cluster
expansion (ACE)

Animation by Will Baldwin
(Small 20, 2303565, 2024)

Class Outline

Crystal Representations

A. Compositional

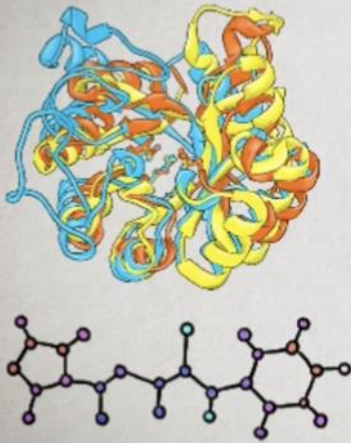
B. Structural

C. Graphs

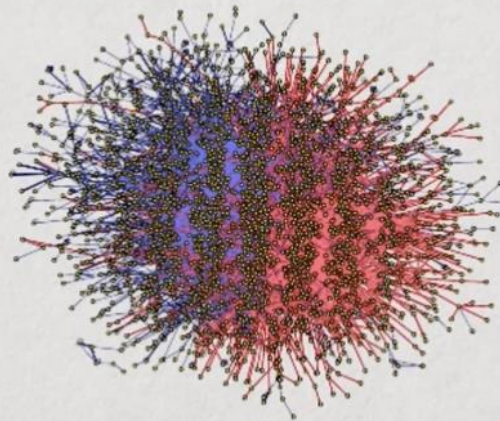
Graphs

Graphs are a representation common to many domains and problems

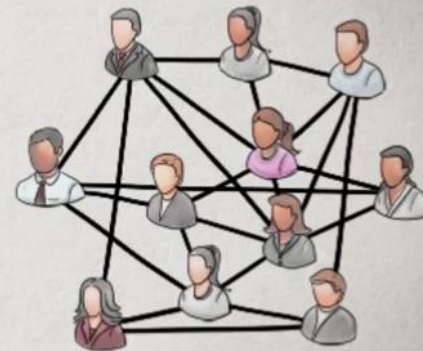
Graphs = systems of relations and interactions



Molecules



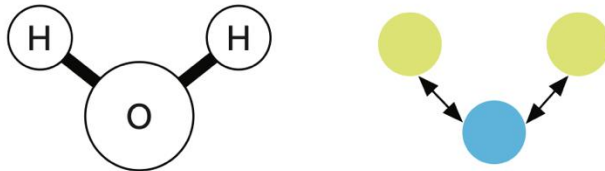
Interactomes



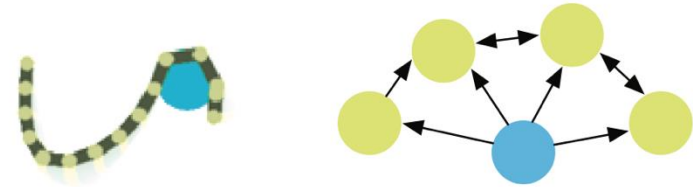
Social networks

Graphs

(a) Molecule



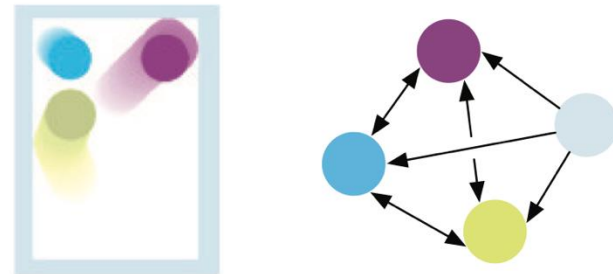
(b) Mass-Spring System



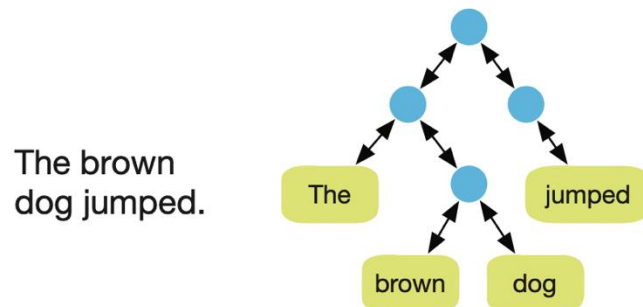
(c) n -body System



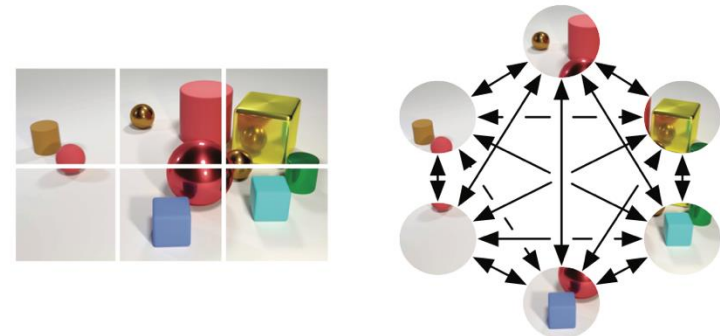
(d) Rigid Body System



(e) Sentence and Parse Tree

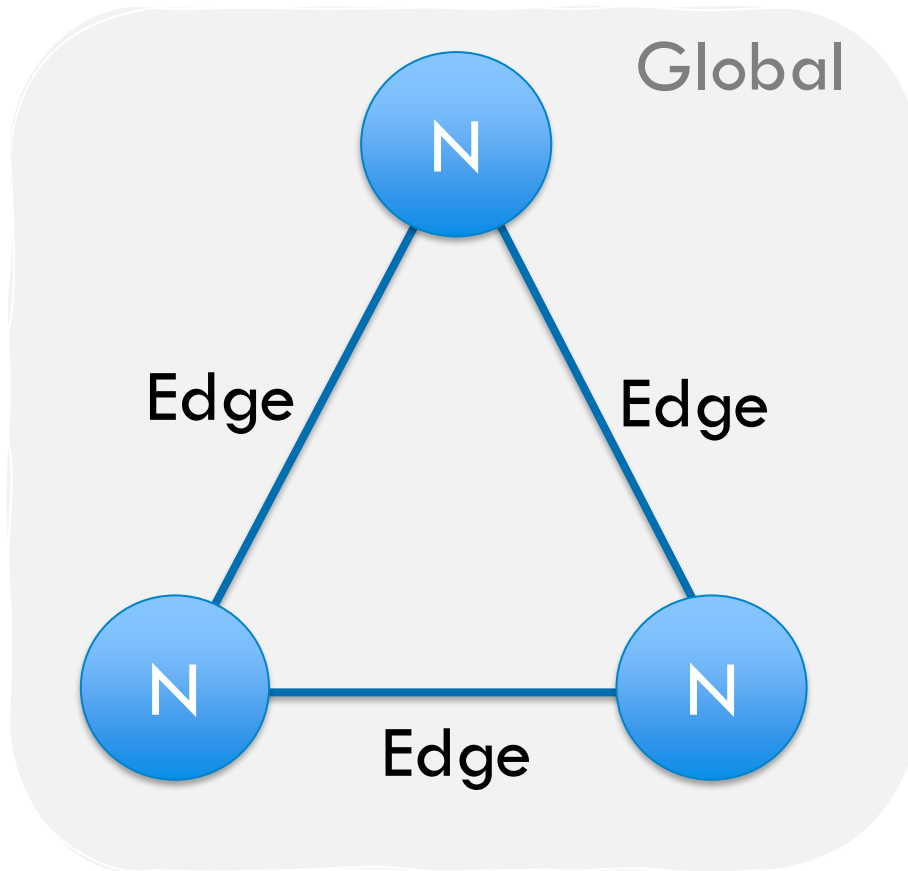


(f) Image and Fully-Connected Scene Graph



Graph Components

Nodes (Vertices), **E**des, **G**lobal Attributes



Crystal systems

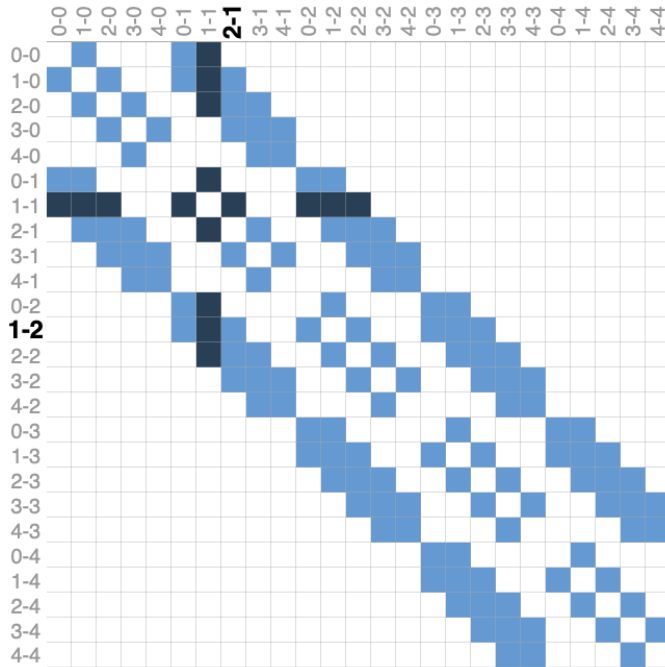
N – atoms

E – bonds

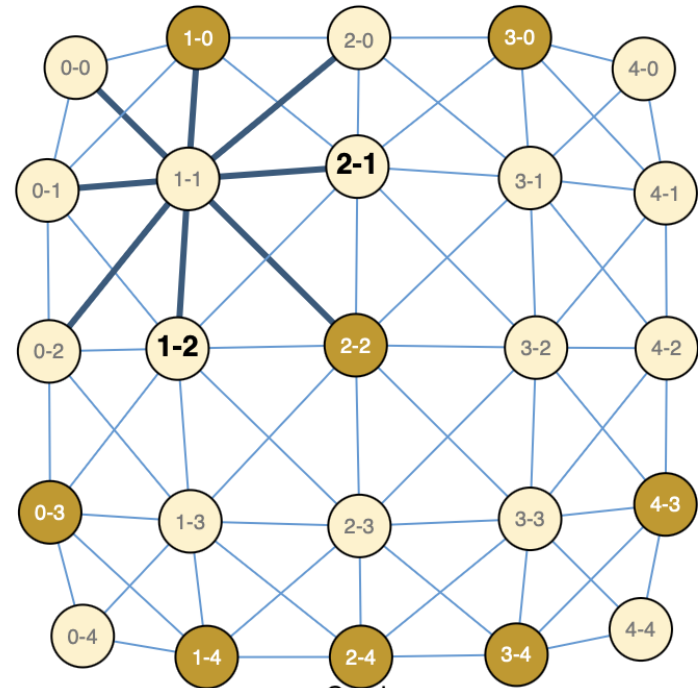
G – unit cell or
materials properties

Graph Components

Nodes (Vertices), Edges, Global Attributes



Adjacency Matrix

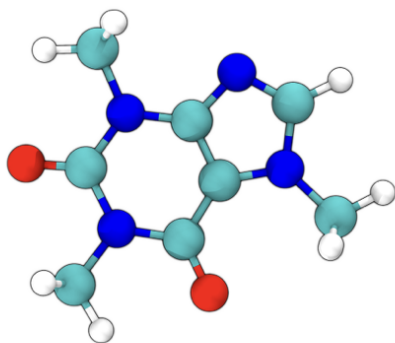


Graph

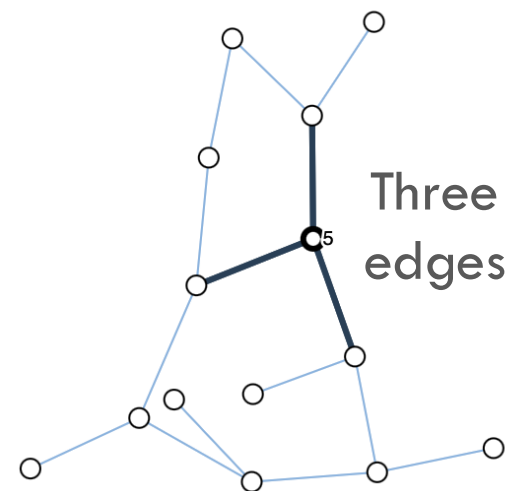
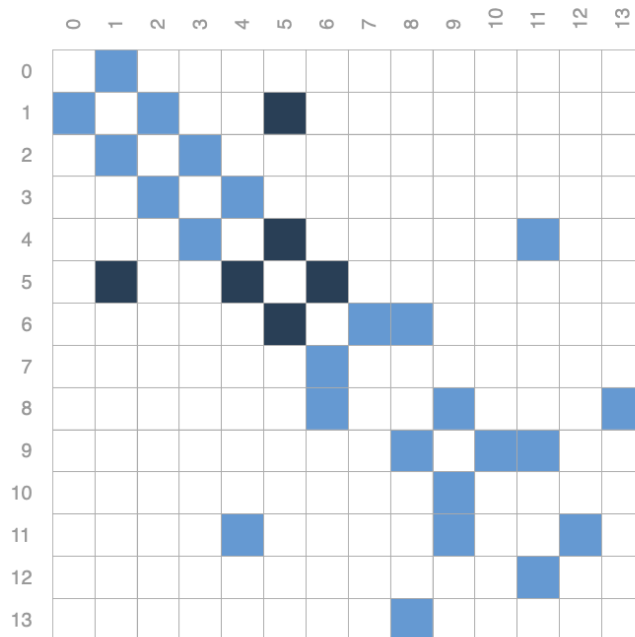
Graphs can be fully connected (every node connected to every other node), but sparse connections are often used

Graph Components

Nodes (or Vertices), Edges, Global Attributes



Molecule
(including H)

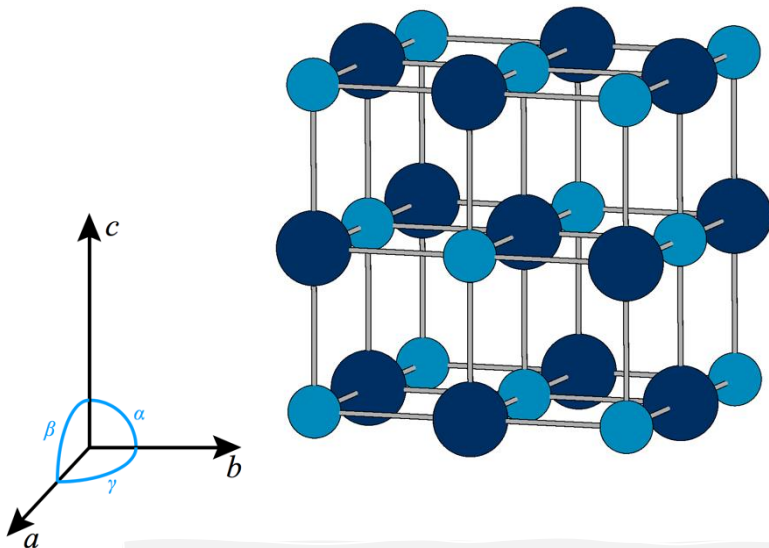


Graph
(excluding H nodes)

For chemical problems, nearest-neighbour connectivity is common, as used in “ball and stick” representations

Crystal Graphs

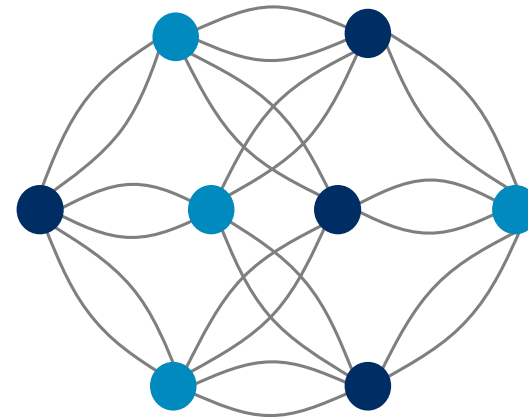
Standard crystallographic representation of materials



Fractional positions xyz of atoms within a unit cell formed of lattice vectors abc

Effective for humans

Crystal graph representation

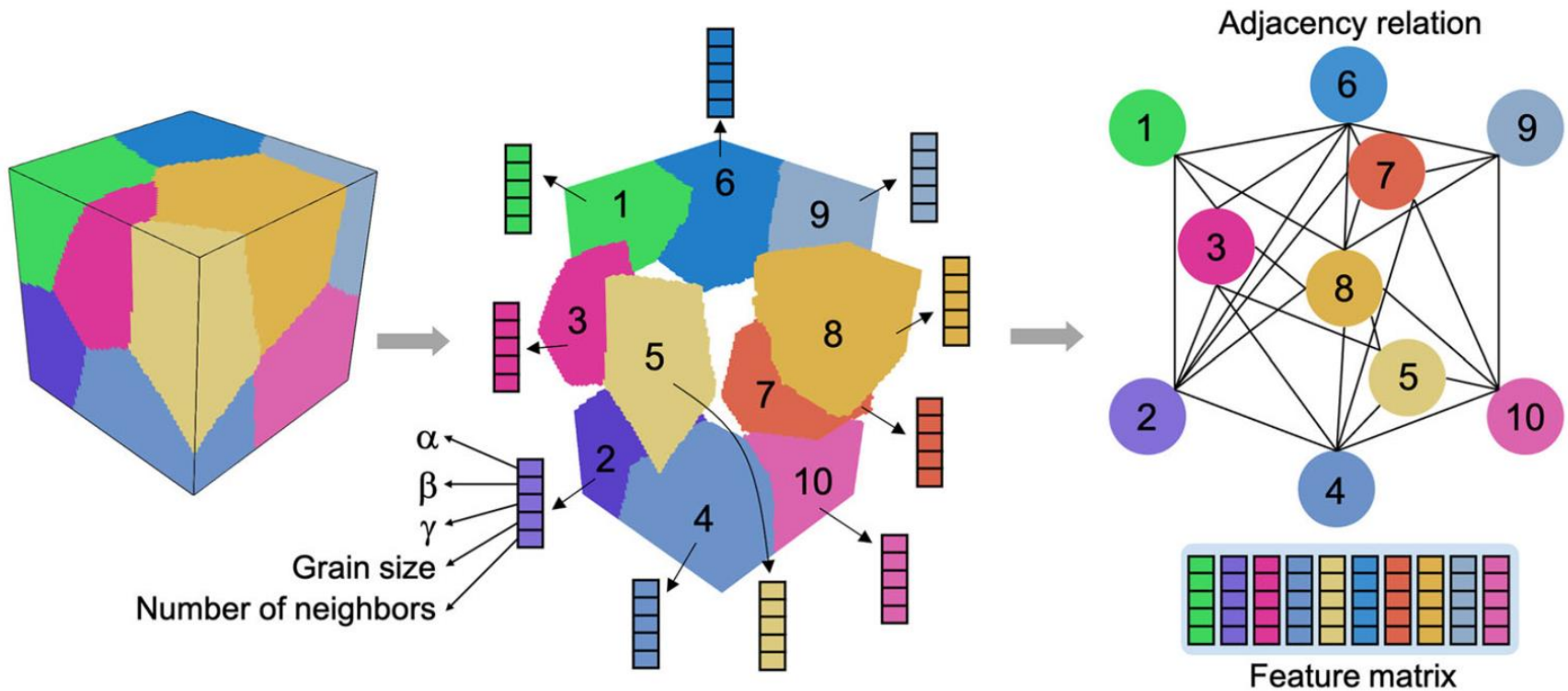


Nodes (atoms) connected by edges (bonds). Multiple edges can describe periodicity

Effective for ML models

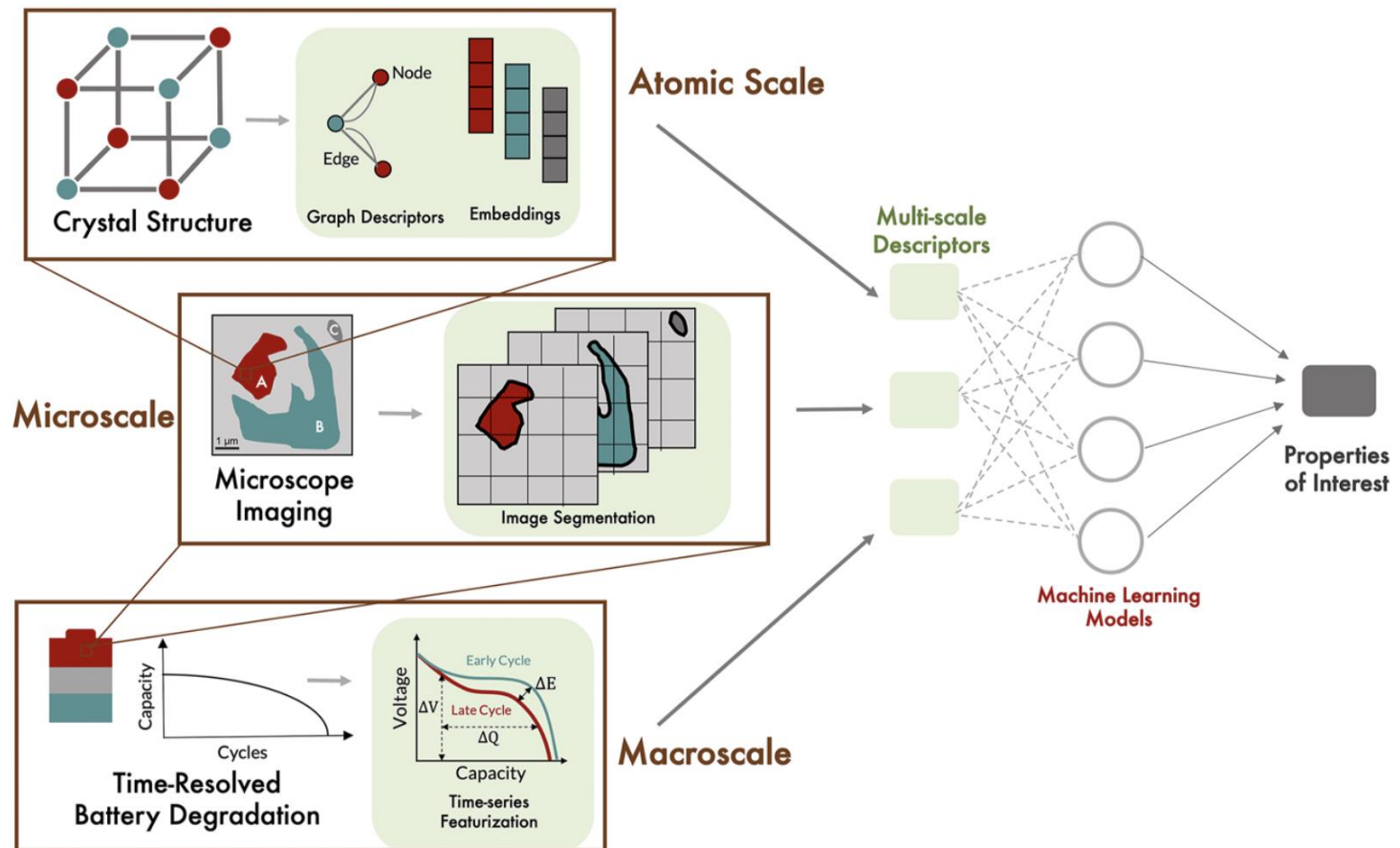
Materials Graphs

Nodes can be used to represent larger structural units of a crystal or even entire grains



Multi-Scale Representations

Ongoing efforts to combine features that bridge from the micro to macroscale; from atoms to devices



Class Outcomes

1. Describe the ways that chemical composition can expanded into vectors
2. Explain how the structure of a material can be represented for machine learning
3. Consider the limitations of a graph-based description of a three-dimensional structure

Activity:

Navigating crystal space
